

## A COMPARISON OF THE SIGNIFICANCE LEVELS AND POWER FUNCTIONS OF SEVERAL SOLUTIONS TO THE MULTIVARIATE BEHRENS-FISHER PROBLEM

**N. de la Rey:** Department of Mathematical Statistics, University of the Orange Free State, P.O. Box 339, Bloemfontein, 9300,  
**and**  
**D.G. Nel:** Republic of South Africa

---

**Key words:** Approximate solutions; Behrens-Fisher problem; comparison; critical values; Monte Carlo studies; multivariate; power function.

---

**Summary:** Nel and Van der Merwe (1986) proposed a new approximate solution to the multivariate Behrens-Fisher problem. In this paper their solution is compared with other solutions to this problem on the basis of Monte Carlo studies with respect to their approximate levels of significance and their power functions.

---

### 1. INTRODUCTION

The Behrens-Fisher problem is posed by the comparison of the means of two populations on the basis of two independent samples when the variances (or covariance matrices in the case of multivariate analysis) are unequal and unknown.

The univariate problem was first studied by Behrens in 1929 and his solution was presented by Fisher in terms of the fiducial theory in 1935. Since then much work has been done and many solutions have been developed, amongst others the approximate degrees of freedom solution and an asymptotic series solution by Welch (1936, 1947).

Some of these univariate solutions were extended to the multivariate analogues, namely:

Bennett (1951), who extended Scheffé's (1943) solution;  
 James (1954), who extended the Welch (1947) series solution;  
 Yao (1965), who gave an extension of the Welch (1936) approximate degrees of freedom solution; and  
 Nel and Van der Merwe (1986), who generalized the Welch (1947) solution by comparing the covariance structures of the exact and approximate distributions.

The purpose of this paper is to compare the relative merits of these four solutions to the multivariate Behrens-Fisher problem on the basis of the following criteria:

- (a) Monte Carlo studies of the levels of significance in section 3, and
- (b) Monte Carlo studies of the power functions of the solutions in section 4.

In this sense the paper is an extension of the work of Subrahmaniam and Subrahmaniam (1973).

## 2 SOLUTIONS TO THE MULTIVARIATE BEHRENS-FISHER PROBLEM

In this section we briefly discuss the underlying theory and resulting test statistic for each of the solutions which we will compare.

### 2.1 Notation

Consider two random samples  $\underline{X}_{ij}$  ( $j = 1, 2, \dots, N_i$ ;  $i = 1, 2$ ) from two independent  $p$ -variate normal distributions  $N_p(\underline{\mu}_i, \underline{\Sigma}_i)$ , where  $\underline{\Sigma}_1 \neq \underline{\Sigma}_2$ . The sample mean variate vectors are independently normally distributed with mean vectors  $\underline{\mu}_i$  and covariance matrices  $\frac{1}{N_i} \underline{\Sigma}_i$ , respectively, indicated as  $\bar{\underline{X}}_i \sim N_p(\underline{\mu}_i, \frac{1}{N_i} \underline{\Sigma}_i)$ ,  $i = 1, 2$ . If the sample covariance matrices are denoted as  $\underline{S}_i$ , then the matrices  $\underline{A}_i = n_i \underline{S}_i$  are independently

distributed as Wishart distributions with parameter matrices  $\Sigma_i$  and degrees of freedom  $n_i = N_i - 1$ , indicated as  $A_i = n_i S_i \sim W_p(n_i, \Sigma_i)$ ,  $i=1, 2$ .

The hypothesis to be tested is  $H_0 : \mu = 0$ , where  $\mu = \mu_1 - \mu_2$ . The maximum likelihood estimate for  $\mu$  is given by  $\underline{Y} = \bar{X}_1 - \bar{X}_2$ , where  $\underline{Y}$  is distributed as

$$(2.1.1) \quad N_p(\underline{\mu}, \Sigma), \quad \Sigma = \frac{1}{N_1} \Sigma_1 + \frac{1}{N_2} \Sigma_2.$$

This notation will be used in all tests and simulations, unless stated otherwise.

## 2.2 Bennett's solution

Bennett (1951) based his test on a multivariate analogue of Scheffé's solution (1943) in the univariate case. The well-known test is an exact one, being directly related to the generalized  $T^2$  of Hotelling.

If the sample of lesser size is denoted as the first sample - so that  $N_1 < N_2$ , the test statistic suggested by Bennett is:

$$T^2 = N_1 \underline{Y}' \mathbf{S}^{-1} \underline{Y},$$

where  $(N_1 - 1)\mathbf{S}$

$$= \sum_{i=1}^{N_1} \left[ x_{i1} - \bar{x}_1 - \sqrt{\frac{N_1}{N_2}} \left[ x_{i2} - \frac{1}{N_1} \sum_{i=1}^{N_1} x_{j2} \right] \right] \left[ x_{i1} - \bar{x}_1 - \sqrt{\frac{N_1}{N_2}} \left[ x_{i2} - \frac{1}{N_1} \sum_{i=1}^{N_1} x_{j2} \right] \right]'.$$

The hypothesis will be rejected if  $\frac{T^2}{N_1 - 1} \cdot \frac{N_1 - p}{p} \geq F_{p; N_1 - p; 1 - \alpha}$ , where  $F_{v_1, v_2; 1 - \alpha}$  denotes the  $100(1 - \alpha)$  percentile of the  $F$ -distribution with  $v_1$  and  $v_2$  degrees of freedom.

### 2.3 James's first-order asymptotic series solution

James (1954) has extended the univariate Welch asymptotic series solution (1947) to a multivariate analogue. Using Taylor expansions, he showed that the critical region is of the form

$$\underline{Y}' \mathbf{S}^{-1} \underline{Y} \geq h_1 \left[ \frac{1}{N_1} \mathbf{S}_1 ; \frac{1}{N_2} \mathbf{S}_2 ; 1 - \alpha \right], \text{ where } \mathbf{S} = \frac{1}{N_1} \mathbf{S}_1 + \frac{1}{N_2} \mathbf{S}_2 \text{ and}$$

where  $h_1$  is to be derived so that

$$P \left[ \underline{Y}' \mathbf{S}^{-1} \underline{Y} \geq h_1 \left[ \frac{1}{N_1} \mathbf{S}_1 ; \frac{1}{N_2} \mathbf{S}_2 ; 1 - \alpha \right] \right] = \alpha + O(n_i^{-2}).$$

He found that  $h_1 \left[ \frac{1}{N_1} \mathbf{S}_1 ; \frac{1}{N_2} \mathbf{S}_2 ; 1 - \alpha \right] = \chi_{p, 1-\alpha}^2 \left[ 1 + \frac{1}{2} \left[ \frac{k_1}{p} + \frac{k_2 \chi_{p, 1-\alpha}^2}{p(p+2)} \right] \right]$  where

$$k_1 = \sum_{i=1}^2 \left[ \text{tr} \mathbf{S}^{-1} \left[ \frac{1}{N_i} \mathbf{S}_i \right] \right]^2 / n_i \text{ and where } \chi_{p, 1-\alpha}^2$$

denotes the  $100(1-\alpha)$ -th percentile of the chi-square distribution with  $p$  degrees of freedom. The value of  $k_2$  is given by

$$k_2 = \sum_{i=1}^2 \left[ \left[ \text{tr} \mathbf{S}^{-1} \left[ \frac{1}{N_i} \mathbf{S}_i \right] \right]^2 + 2 \text{tr} \left[ \mathbf{S}^{-1} \left[ \frac{1}{N_i} \mathbf{S}_i \right] \right]^2 \right] / n_i.$$

This value  $h_1 \left[ \frac{1}{N_1} \mathbf{S}_1 ; \frac{1}{N_2} \mathbf{S}_2 ; 1 - \alpha \right]$  is the critical value of a test of approximate size  $\alpha$  and the hypothesis  $H_0 : \mu_1 = \mu_2$  will be rejected if

$$T^2 = \underline{Y}' \mathbf{S}^{-1} \underline{Y} \geq h_1 \left[ \frac{1}{N_1} \mathbf{S}_1 ; \frac{1}{N_2} \mathbf{S}_2 ; 1 - \alpha \right].$$

## 2.4 Yao's approximate degrees of freedom solution

Yao (1965) studied an extension of the univariate Welch (1936) approximate degrees of freedom solution and suggested a multivariate generalization based on the  $T^2$ -distribution of Hotelling.

In constructing the test with approximate significance level  $\alpha$ , she considered the test statistic  $\underline{Y}'\underline{S}^{-1}\underline{Y}$  and assumed  $n\underline{S}$  to be distributed approximately Wishart with unknown degrees of freedom  $n$  and parameter matrix  $\underline{\Sigma}$  as in (2.1.1).

For an arbitrary constant vector  $\underline{b}$  it then follows that  $\underline{b}'\underline{Y} \sim N(0, \underline{b}'\underline{\Sigma}\underline{b})$  independently of  $n\underline{b}'\underline{S}\underline{b} \sim W(n, \underline{b}'\underline{\Sigma}\underline{b})$ .

Thus  $n(\underline{b}'\underline{S}\underline{b}) \sim (\underline{b}'\underline{\Sigma}\underline{b})\chi_n^2$ , and  $w_b = \frac{(\underline{b}'\underline{Y})^2}{(\underline{b}'\underline{S}\underline{b})} \sim t_n^2$ .

Yao (1965) shows that  $w_b$  is maximized if  $\underline{b} = \underline{S}^{-1}\underline{Y}$ . Applying this supremum over all  $\underline{b}$  she hopes that:

$$P(\underline{Y}'\underline{S}^{-1}\underline{Y} \geq T_{p,n;1-\alpha}^2) = \alpha + o(n_i^{-2})$$

where for the maximizing value of  $\underline{b}$ :  $w_b = \underline{Y}'\underline{S}^{-1}\underline{Y} \sim T_{p,n}^2$ . By  $T_{p,n;1-\alpha}^2$  we will denote  $(np/(n-p+1))F_{p,n-p+1;1-\alpha}$ .

The approximate degrees of freedom  $f_Y$  is then the reciprocal of

$$\left[ \left[ \underline{b}' \left[ \frac{1}{N_1} \underline{S}_1 \right] \underline{b} \right] / (\underline{b}'\underline{S}\underline{b}) \right]^2 / n_1 + \left[ \left[ \underline{b}' \left[ \frac{1}{N_2} \underline{S}_2 \right] \underline{b} \right] / (\underline{b}'\underline{S}\underline{b}) \right]^2 / n_2.$$

Let  $F = \frac{T^2}{f_Y} \cdot \frac{f_Y - p + 1}{p}$ , where  $T^2 = \underline{Y}'\underline{S}^{-1}\underline{Y}$ , then the null hypothesis will

be rejected with a test of approximate size  $\alpha$  if  $F \geq F_{p, f_Y - p + 1; 1 - \alpha}$ .

## 2.5 Nel and Van der Merwe's (NAV) solution

Nel and Van der Merwe (1986) presented a method which is similar to the method used by Welch (1947) in the univariate case. In the Behrens-Fisher situation we have that  $\underline{Y}$  is distributed independently of  $\mathbf{S} = \frac{1}{N_1} \mathbf{S}_1 + \frac{1}{N_2} \mathbf{S}_2$ . This matrix  $\mathbf{S}$  has a "sum of Wisharts" distribution, denoted as

$$\mathbf{S} \sim SOW(N_1 - 1, N_2 - 1; g_1 \Sigma_1, g_2 \Sigma_2),$$

where  $g_i = \frac{1}{N_i(N_i-1)}$ ,  $i = 1, 2$ . This distribution is described in detail by Nel and Van der Merwe (1986). Using an approach suggested by Tan and Gupta (1983) the distribution is approximated by a Wishart distribution

$$U \sim W_p\left[f_n, \frac{1}{f_n} \left[ \frac{1}{N_1} \Sigma_1 + \frac{1}{N_2} \Sigma_2 \right]\right] = W_p\left[f_n, \frac{1}{f_n} \Sigma\right]$$

and the degrees of freedom  $f_n$  is approximated by:

$$(2.5.1) \quad f = \frac{\text{tr}\left[\frac{1}{N_1}\mathbf{S}_1 + \frac{1}{N_2}\mathbf{S}_2\right]^2 + \text{tr}^2\left[\frac{1}{N_1}\mathbf{S}_1 + \frac{1}{N_2}\mathbf{S}_2\right]}{\frac{1}{N_1-1}\left[\text{tr}\left[\frac{1}{N_1}\mathbf{S}_1\right]\right]^2 + \text{tr}^2\left[\frac{1}{N_1}\mathbf{S}_1\right] + \frac{1}{N_2-1}\left[\text{tr}\left[\frac{1}{N_2}\mathbf{S}_2\right]\right]^2 + \text{tr}^2\left[\frac{1}{N_2}\mathbf{S}_2\right]}$$

Here  $\text{tr}^2 \mathbf{A}$  denotes  $(\text{tr} \mathbf{A})^2$ . Let  $F = \frac{T^2}{f} \cdot \frac{f-p+1}{p}$ , where  $T^2 = \underline{Y}' \mathbf{S}^{-1} \underline{Y}$ , then the null hypothesis will be rejected with a test of approximate size  $\alpha$  if  $F \geq F_{p, f-p+1; 1-\alpha}$ .

### 3. MONTE CARLO STUDIES OF THE LEVEL OF SIGNIFICANCE OF EACH TEST

A Monte Carlo technique was used for the determination of  $\alpha$ , the level of significance of the tests of James, Yao and Nel and Van der Merwe. As Bennett's method achieves the exact significance level  $\alpha$ , it is not included here.

#### 3.1 Brief description of the method used

For a given pair of positive definite symmetric matrices,  $\Sigma_1$  and  $\Sigma_2$ , there exists a non-singular matrix  $C$  and a diagonal matrix  $\Lambda$ , with elements

$$0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_p \leq 1$$

such that  $C\Sigma C' = I_p$ , where  $\Sigma = \frac{1}{N_1}\Sigma_1 + \frac{1}{N_2}\Sigma_2$ .

Let  $C\left[\frac{1}{N_1}\Sigma_1\right]C' = \Lambda_1$  and  $C\left[\frac{1}{N_2}\Sigma_2\right]C' = \Lambda_2$ , where  $\Lambda_1 = \Lambda$  and  $\Lambda_2 = I_p - \Lambda$ .

The significance level of a test for a given set  $\left[\frac{1}{N_1}\Sigma_1, \frac{1}{N_1}\Sigma_2\right]$  is a function of the elements of the diagonal matrix  $\Lambda$  only, and may be written as

$$\pi(\Lambda) = P(\underline{Y}'S^{-1}\underline{Y} \geq C_{1-\alpha}) = P(\underline{z}'M^{-1}\underline{z} \geq C_{1-\alpha}),$$

where  $C_{1-\alpha}$  is the critical value of the test.

This equality must hold true, since if  $C$  is a matrix such that  $C\Sigma C' = I_p$

$$\text{then } \underline{Y}'S^{-1}\underline{Y} = \underline{z}'M^{-1}\underline{z}, \quad \text{where } M = CSC' = M_1 + M_2$$

$$\text{with } M_i = C\left[\frac{1}{N_i}S_i\right]C' \quad \text{and} \quad \underline{z} = C\underline{Y} \sim N(0, I_p)$$

when the null hypothesis is true.

In particular, the significance levels of the three tests are as follows:

1. Yao

The approximate degrees of freedom  $f_Y$  is given by

$$\frac{1}{f_Y} = \frac{1}{n_1} \left[ \frac{\underline{z}' \mathbf{M}^{-1} \mathbf{M}_1 \mathbf{M}^{-1} \underline{z}}{\underline{z}' \mathbf{M}^{-1} \underline{z}} \right]^2 + \frac{1}{n_2} \left[ \frac{\underline{z}' \mathbf{M}^{-1} \mathbf{M}_2 \mathbf{M}^{-1} \underline{z}}{\underline{z}' \mathbf{M}^{-1} \underline{z}} \right]^2.$$

and the significance level by

$$(3.1.1) \quad \alpha_Y(\Lambda) = P \left[ \underline{z}' \mathbf{M}^{-1} \underline{z} \geq T_{p; f_Y; 1-\alpha}^2 \right].$$

2. James

The critical value of the test is given by

$$h_1(\mathbf{M}_1; \mathbf{M}_2; 1 - \alpha) = \chi_{p; 1-\alpha}^2 \left[ 1 + \frac{1}{2} \left[ \frac{k_1}{p} + \frac{k_2 \chi_{p; 1-\alpha}^2}{p(p+2)} \right] \right],$$

where 
$$k_1 = \sum_{i=1}^2 \frac{(tr \mathbf{M}^{-1} \mathbf{M}_i)^2}{n_i}, \quad n_i = N_i - 1$$

and 
$$k_2 = \sum_{i=1}^2 \frac{(tr \mathbf{M}^{-1} \mathbf{M}_i)^2 + 2 tr(\mathbf{M}^{-1} \mathbf{M}_i)^2}{n_i}.$$

Thus the significance level is:

$$(3.1.2) \quad \alpha_J(\Lambda) = P(\underline{z}' \mathbf{M}^{-1} \underline{z} \geq h_1(\mathbf{M}_1; \mathbf{M}_2; 1 - \alpha)).$$

### 3. Nel and Van der Merwe

Their approximate degrees of freedom  $f$  is given by (2.5.1) with  $S_1$  and  $S_2$  replaced by  $M_1$  and  $M_2$  and the significance level is:

$$(3.1.3) \quad \alpha_N(\Lambda) = P(\underline{z}' \mathbf{M}^{-1} \underline{z} \geq T_{p;f;1-\alpha}^2).$$

### 3.2 Numerical calculations

To simplify numerical calculations, expressions (3.1.1), (3.1.2) and (3.1.3) may be written as mathematical expectations

$$\alpha_Y(\Lambda) = E(Q(R_Y)), \quad \alpha_J(\Lambda) = E(Q(R_J)) \quad \text{and} \quad \alpha_N(\Lambda) = E(Q(R_N)),$$

where

$$(3.2.1) \quad R_Y = \frac{nT_{p;f_Y;1-\alpha}^2}{(\underline{d}' \mathbf{M}^{-1} \underline{d})w}$$

$$(3.2.2) \quad R_J = \frac{nh_1(\mathbf{M}_1; \mathbf{M}_2; 1 - \alpha)}{(\underline{d}' \mathbf{M}^{-1} \underline{d})w}$$

and

$$(3.2.3) \quad R_N = \frac{nT_{p;f;1-\alpha}^2}{(\underline{d}' \mathbf{M}^{-1} \underline{d})w}$$

with  $\underline{d} = \frac{\underline{z}}{\sqrt{\underline{z}' \underline{z}}}$  and  $w = \sum_{i=1}^2 n_i \text{tr}(\Lambda_i^{-1} \mathbf{M}_i)$ . Here  $Q(a)$  denotes the

complement of the cumulative distribution function of a central  $F$  variable with  $p$  and  $np$  degrees of freedom, where  $n = n_1 + n_2$ .

### 3.3 The practical implementation

For any given set of parameters  $(n_1, n_2, \Lambda)$  the values  $R_Y$ ,  $R_J$  and  $R_N$  may be generated from a set of standard variables  $z$ ,  $W_1$  and  $W_2$  where

$$z: p \times 1 \sim N(0, I_p), \quad W_i \sim W_p(n_i, I_p) \quad \text{and} \quad M_i = \frac{1}{n_i} \left[ \Lambda_i^{\frac{1}{2}} W_i \Lambda_i^{\frac{1}{2}} \right] \quad (i = 1, 2).$$

When simulated values of  $z$ ,  $W_1$  and  $W_2$  are available,  $R_Y$ ,  $R_J$  and  $R_N$  may be computed. The resulting estimates  $\alpha_Y$ ,  $\alpha_J$  and  $\alpha_N$  are correlated and provide a closer comparison of the three solutions.

If  $N$  independent sets of  $z$ ,  $W_1$  and  $W_2$  are simulated,  $N$  sets of values of  $R_Y$ ,  $R_J$  and  $R_N$  would be available for each  $\Lambda$ . As estimates for the significance levels  $\alpha_Y$ ,  $\alpha_J$  and  $\alpha_N$ , the following may be used.

$$\bar{\alpha}_T(\Lambda) = \frac{1}{N} \sum_{i=1}^N Q(R_{Ti}), \quad \text{where } T = Y, J \text{ or } N.$$

We used 500 sets of  $z$ ,  $W_1$ , and  $W_2$  for  $p = 2$ , gradually increasing this number for larger values of  $p$ . For  $p = 10$  two thousand samples were used. The results obtained were verified by comparing them to those obtained by Subrahmaniam and Subrahmaniam (1973), as well as by using the case where all eigenvalues  $\lambda_i$  are equal to  $\frac{n_i}{n} (i = 1, \dots, p)$ .

The results of the computations are presented in table 1.

It seems that the NAV solution tends to give the smallest significance levels. However, in a few cases smaller levels were obtained with Yao's solution, whilst in others both solutions gave exactly the same values. Except for the cases where the eigenvalues are combinations of 0.1 and 0.9, James's solution invariably yields the largest approximate significance levels.

Table 1

Monte Carlo estimates of the level of significance of each test. Nominal level  $\alpha = 0,05$

For dimensionality:  $p = 2$

| $n_1$ | $n_2$ | $\lambda_1$ | $\lambda_2$ | Yao   | James | NAV   |
|-------|-------|-------------|-------------|-------|-------|-------|
| 6     | 12    | 0,1         | 0,1         | 0,051 | 0,057 | 0,049 |
|       |       | 0,33        | 0,33        | 0,047 | 0,053 | 0,048 |
|       |       | 0,25        | 0,75        | 0,045 | 0,052 | 0,044 |
|       |       | 0,1         | 0,9         | 0,051 | 0,062 | 0,053 |
|       |       | 0,9         | 0,9         | 0,063 | 0,075 | 0,047 |
| 12    | 12    | 0,1         | 0,1         | 0,052 | 0,057 | 0,049 |
|       |       | 0,5         | 0,5         | 0,046 | 0,049 | 0,046 |
|       |       | 0,25        | 0,75        | 0,046 | 0,050 | 0,046 |
|       |       | 0,1         | 0,9         | 0,048 | 0,053 | 0,048 |
|       |       | 0,9         | 0,9         | 0,054 | 0,059 | 0,051 |
| 6     | 18    | 0,1         | 0,1         | 0,049 | 0,052 | 0,048 |
|       |       | 0,25        | 0,25        | 0,048 | 0,052 | 0,048 |
|       |       | 0,25        | 0,75        | 0,048 | 0,053 | 0,046 |
|       |       | 0,1         | 0,9         | 0,053 | 0,061 | 0,056 |
|       |       | 0,9         | 0,9         | 0,060 | 0,071 | 0,044 |

For dimensionality:  $p = 3$

| $n_1$ | $n_2$ | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | Yao   | James | NAV   |
|-------|-------|-------------|-------------|-------------|-------|-------|-------|
| 6     | 12    | 0,1         | 0,1         | 0,1         | 0,050 | 0,063 | 0,046 |
|       |       | 0,33        | 0,33        | 0,33        | 0,048 | 0,062 | 0,050 |
|       |       | 0,25        | 0,5         | 0,75        | 0,048 | 0,060 | 0,043 |
|       |       | 0,1         | 0,1         | 0,9         | 0,047 | 0,063 | 0,048 |
|       |       | 0,9         | 0,9         | 0,9         | 0,086 | 0,103 | 0,036 |
| 12    | 12    | 0,1         | 0,1         | 0,1         | 0,046 | 0,056 | 0,039 |
|       |       | 0,25        | 0,5         | 0,75        | 0,046 | 0,054 | 0,047 |
|       |       | 0,5         | 0,5         | 0,5         | 0,049 | 0,057 | 0,050 |
|       |       | 0,1         | 0,1         | 0,9         | 0,046 | 0,056 | 0,045 |
|       |       | 0,9         | 0,9         | 0,9         | 0,054 | 0,064 | 0,046 |
| 6     | 18    | 0,1         | 0,1         | 0,1         | 0,049 | 0,056 | 0,048 |
|       |       | 0,25        | 0,25        | 0,25        | 0,046 | 0,053 | 0,047 |
|       |       | 0,25        | 0,5         | 0,75        | 0,048 | 0,055 | 0,042 |
|       |       | 0,1         | 0,1         | 0,9         | 0,045 | 0,057 | 0,049 |
|       |       | 0,9         | 0,9         | 0,9         | 0,094 | 0,108 | 0,038 |

These values get worse for large eigenvalues, especially when  $n_1$  is much smaller than  $n_2$ . Quite the opposite seems to happen with the NAV solution. For example, if  $p = 3$ ,  $\lambda_1 = \lambda_2 = \lambda_3 = 0.9$ ,  $n_1 = 6$  and  $n_2 = 12$ , we found the estimated significance levels to be:

Yao: 0,086; James: 0,103 and NAV: 0,036.

For such large eigenvalues, the significance levels of Yao's solution also tend to increase, while the NAV probabilities are always smaller.

When the dimensionality,  $p$ , is increased, the estimated significance levels of the Yao and the NAV solutions do not change much and stay relatively small. The same cannot be said of the series solution of James, where the values obtained become larger as the dimensionality increases.

#### 4. MONTE CARLO STUDIES OF THE POWER OF EACH TEST

The same technique used in the previous section to estimate significance levels can now be used to determine the power of the tests of James, Yao and the NAV solution, independent of eigenvalues as a result of the transformation on the covariance matrices. The power of this test is thus computed directly.

##### 4.1 The power function using a Monte Carlo estimation technique

Following the procedure for obtaining the representation of  $\alpha$ , the power function can be written as the cumulative distribution function of a noncentral  $F$  variable.

Consider, for example, Yao's approximate degrees of freedom method, with power function:

$$\begin{aligned}\Pi_Y(\phi) &= P(\text{Reject } H_0 | \phi), \text{ where } \phi = \mu' \Sigma^{-1} \mu \\ &= P\left(\mathbf{z}' \mathbf{M}^{-1} \mathbf{z} \geq T_{p; f_Y, 1-\alpha}^2\right) \\ &= P(X \geq R_Y), \quad X \sim F_{p; n_p}(\phi) \\ &= E[Q^*(R_Y)]\end{aligned}$$

where  $Q^*(a)$  is the complement of the cumulative distribution function of a noncentral  $F$  variable with  $p$  and  $np$  degrees of freedom and noncentrality parameter  $\phi$ . To simplify numerical calculations, the following relation between the noncentral  $f$ -distribution and noncentral Beta Type I distribution is useful.

If  $F \sim F_{n,m}(\phi)$ , then  $B = \frac{nF}{M + nF} \sim \beta_1\left[\frac{n}{2}; \frac{m}{2}; \frac{\phi}{2}\right]$ .

Now,

$$E[Q^*(R_Y)] = E[P(X \geq R_Y | X \sim F_{p,np}(\phi))]$$

and

$$P(X > R_Y | X \sim F_{p,np}(\phi)) = P\left[\frac{X \cdot p}{np + Xp} > R_Y^* | X \sim F_{p,np}(\phi)\right], \text{ where}$$

$$\begin{aligned} R_Y^* &= \frac{R_Y}{n + R_Y} \\ &= P\left[B > R_Y^* | B \sim \beta_1\left[\frac{p}{2}; \frac{np}{2}; \frac{\phi}{2}\right]\right] \\ &= \sum_{k=0}^{\infty} \text{Poisson}\left[k, \frac{\phi}{2}\right] P\left[B > R_Y^* | B \sim \beta_1\left[\frac{1}{2}(p+2k); \frac{np}{2}\right]\right] \\ &= \sum_{k=0}^{\infty} \frac{e^{-\frac{\phi}{2}} \left(\frac{\phi}{2}\right)^k}{k!} P\left[X > \frac{R_Y^* \cdot np}{(p+2k)(1-R_Y^*)} \mid X \sim F_{p+2k,np}\right]. \end{aligned}$$

Similar representations can be given for the James and NAV procedures.

If  $N$  independent sets of  $z$ ,  $W_1$  and  $W_2$  are generated once again, we will have  $N$  sets of values of  $R_Y$ ,  $R_J$  and  $R_N$  for each  $\Lambda$ .

Now

$$\Pi_Y(\phi) = \frac{1}{N} \sum_{i=1}^N Q^*(R_{Y_i}), \Pi_J(\phi) = \frac{1}{N} \sum_{i=1}^N Q^*(R_{J_i}) \text{ and } \Pi_N(\phi) = \frac{1}{N} \sum_{i=1}^N Q^*(R_{N_i})$$

may be used as estimates for the power of the tests, for a given value of  $\phi$ .

## 4.2 The practical implementation

As before, we varied the number of sets of  $z$ ,  $W_1$  and  $W_2$  that were simulated, between 500 for  $p = 2$  and 2000 for  $p = 10$ . The same combinations of eigenvalues, for a given  $p$  and given sample sizes, were used again. The power of each test was then calculated for six values of the noncentrality parameter. The results obtained are given in table 2. More extensive tables are available. Graphical representations of the power functions of the different solutions for each value of  $p$ , for selected combinations of eigenvalues and sample sizes, are given in figures 1 to 3.

A comparison of our results with those obtained by Subrahmaniam and Subrahmaniam (1973) suggests the possibility of their usage of  $\frac{\phi}{2}$  as noncentrality parameter, instead of  $\phi$ .

## 5. CONCLUSION

In this paper we described four tests which can be used for the Behrens-Fisher situation in multivariate analysis. These tests were compared on the basis of their significance levels and their power functions.

The two solutions which stand out are those of Nel and Van der Merwe, and of Yao. On the basis of significance levels it seems that the NAV solution is the better, while the latter solution gives better power in some cases.

As also noted by Subrahmaniam and Subrahmaniam (1973), the solution of James (1954) seems to have the highest power, but these values should be seen relative to the high significance levels of this test which determine the position of the power curves.

It is rather difficult to judge Bennett's solution as it could only be compared to the others in terms of power. Although his solution achieves the exact level of significance  $\alpha$ , its power is without exception much lower.

On the basis of the criteria investigated it seems that the NAV solution or that of Yao can be used. The NAV solution, however, seems to be better in the cases where the covariance matrices are quite different, i.e. where the eigenvalues are very large.

**Table 2.** Results of Monte Carlo studies of the power of the tests of Yao (Y), James (J), NAV (N) and Bennett (B) for  $\alpha = 0,05$ .

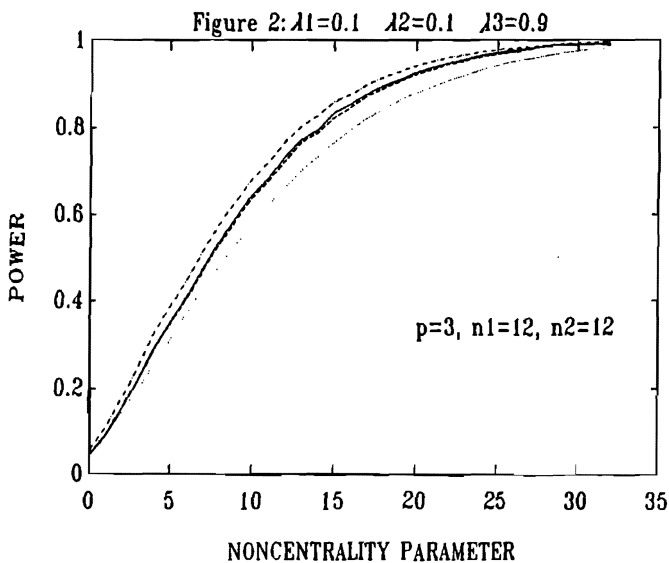
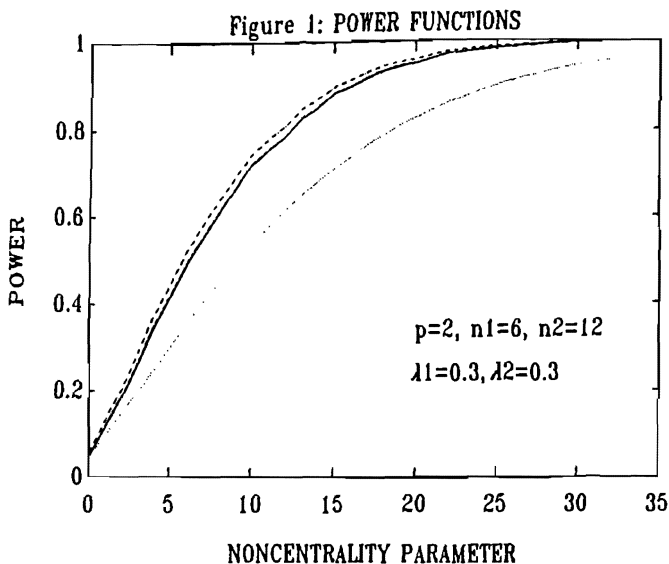
|     |       |       |             |             |             |             |             | Noncentrality parameter $\phi$ |       |       |       |       |       |
|-----|-------|-------|-------------|-------------|-------------|-------------|-------------|--------------------------------|-------|-------|-------|-------|-------|
| $p$ | $n_1$ | $n_2$ | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | $\lambda_4$ | $\lambda_5$ | 1                              | 2     | 4     | 8     | 16    | 32    |
| 2   | 6     | 12    | 0,1         | 0,1         |             |             |             | Y 0,117                        | 0,191 | 0,344 | 0,614 | 0,899 | 0,966 |
|     |       |       |             |             |             |             |             | J 0,129                        | 0,207 | 0,368 | 0,641 | 0,913 | 0,997 |
|     |       |       |             |             |             |             |             | N 0,114                        | 0,187 | 0,339 | 0,610 | 0,898 | 0,996 |
| 2   | 6     | 12    | 0,25        | 0,75        |             |             |             | Y 0,106                        | 0,175 | 0,321 | 0,584 | 0,878 | 0,979 |
|     |       |       |             |             |             |             |             | J 0,120                        | 0,195 | 0,350 | 0,621 | 0,903 | 0,996 |
|     |       |       |             |             |             |             |             | N 0,106                        | 0,174 | 0,320 | 0,568 | 0,882 | 0,987 |
| 2   | 6     | 12    | 0,1         | 0,9         |             |             |             | Y 0,114                        | 0,183 | 0,327 | 0,548 | 0,875 | 0,973 |
|     |       |       |             |             |             |             |             | J 0,132                        | 0,208 | 0,362 | 0,627 | 0,904 | 0,996 |
|     |       |       |             |             |             |             |             | N 0,115                        | 0,184 | 0,327 | 0,586 | 0,879 | 0,985 |
|     |       |       |             |             |             |             |             | B 0,096                        | 0,145 | 0,247 | 0,445 | 0,736 | 0,955 |
|     |       |       |             |             |             |             |             | Y 0,118                        | 0,192 | 0,346 | 0,616 | 0,901 | 0,997 |
| 2   | 12    | 12    | 0,1         | 0,1         |             |             |             | J 0,129                        | 0,207 | 0,368 | 0,642 | 0,915 | 0,997 |
|     |       |       |             |             |             |             |             | N 0,114                        | 0,187 | 0,339 | 0,611 | 0,900 | 0,995 |
|     |       |       |             |             |             |             |             | Y 0,122                        | 0,196 | 0,351 | 0,622 | 0,904 | 0,995 |
| 2   | 12    | 12    | 0,9         | 0,9         |             |             |             | J 0,132                        | 0,212 | 0,374 | 0,648 | 0,917 | 0,998 |
|     |       |       |             |             |             |             |             | N 0,117                        | 0,191 | 0,346 | 0,618 | 0,903 | 0,995 |
|     |       |       |             |             |             |             |             | B 0,112                        | 0,181 | 0,326 | 0,588 | 0,883 | 0,995 |
| 2   | 6     | 18    | 0,1         | 0,1         |             |             |             | Y 0,122                        | 0,200 | 0,369 | 0,650 | 0,923 | 0,988 |
|     |       |       |             |             |             |             |             | J 0,129                        | 0,209 | 0,381 | 0,663 | 0,929 | 0,998 |
|     |       |       |             |             |             |             |             | N 0,122                        | 0,199 | 0,368 | 0,649 | 0,923 | 0,998 |
| 2   | 6     | 18    | 0,25        | 0,75        |             |             |             | Y 0,115                        | 0,186 | 0,333 | 0,617 | 0,892 | 0,988 |
|     |       |       |             |             |             |             |             | J 0,125                        | 0,196 | 0,355 | 0,645 | 0,910 | 0,997 |
|     |       |       |             |             |             |             |             | N 0,113                        | 0,178 | 0,331 | 0,616 | 0,892 | 0,988 |

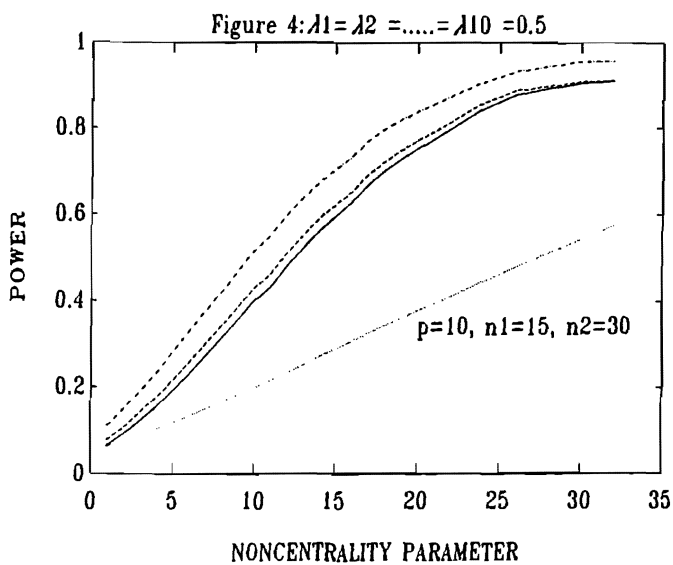
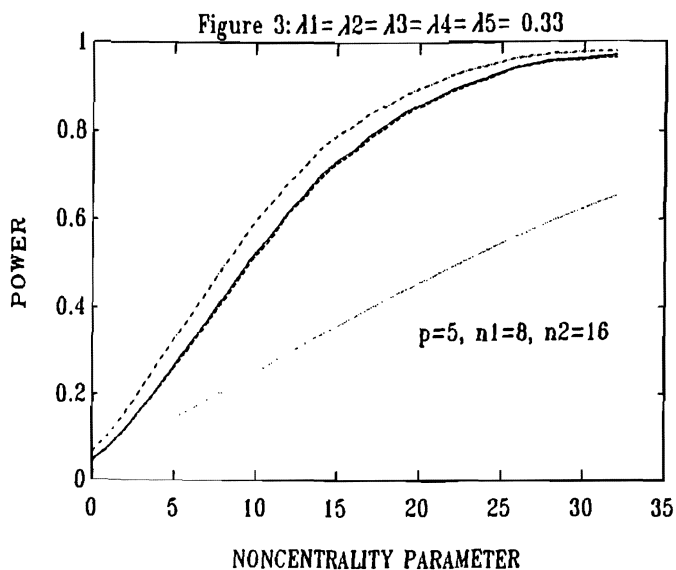
| $p$ | $n_1$ | $n_2$ | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | $\lambda_4$ | $\lambda_5$ |   | 1     | 2     | 4     | 8     | 16    | 32    |
|-----|-------|-------|-------------|-------------|-------------|-------------|-------------|---|-------|-------|-------|-------|-------|-------|
| 3   | 12    | 12    | 0,1         | 0,1         | 0,1         |             |             | B | 0,096 | 0,145 | 0,247 | 0,445 | 0,736 | 0,955 |
|     |       |       |             |             |             |             |             | Y | 0,092 | 0,160 | 0,286 | 0,500 | 0,826 | 0,959 |
|     |       |       |             |             |             |             |             | J | 0,110 | 0,185 | 0,324 | 0,555 | 0,865 | 0,989 |
| 3   | 12    | 12    | 0,1         | 0,1         | 0,9         |             |             | N | 0,082 | 0,146 | 0,268 | 0,485 | 0,820 | 0,962 |
|     |       |       |             |             |             |             |             | Y | 0,095 | 0,165 | 0,278 | 0,529 | 0,858 | 0,987 |
|     |       |       |             |             |             |             |             | J | 0,112 | 0,189 | 0,313 | 0,574 | 0,885 | 0,995 |
| 4   | 12    | 12    | 0,1         | 0,1         | 0,1         | 0,1         |             | N | 0,095 | 0,165 | 0,280 | 0,535 | 0,863 | 0,992 |
|     |       |       |             |             |             |             |             | B | 0,094 | 0,145 | 0,255 | 0,477 | 0,794 | 0,981 |
|     |       |       |             |             |             |             |             | Y | 0,093 | 0,156 | 0,265 | 0,411 | 0,775 | 0,888 |
| 4   | 8     | 16    | 0,25        | 0,5         | 0,5         | 0,75        |             | J | 0,119 | 0,193 | 0,318 | 0,522 | 0,842 | 0,957 |
|     |       |       |             |             |             |             |             | N | 0,072 | 0,127 | 0,226 | 0,403 | 0,753 | 0,872 |
|     |       |       |             |             |             |             |             | B | 0,084 | 0,122 | 0,207 | 0,389 | 0,697 | 0,951 |
| 4   | 8     | 16    | 0,1         | 0,1         | 0,9         | 0,9         |             | Y | 0,088 | 0,137 | 0,255 | 0,468 | 0,795 | 0,950 |
|     |       |       |             |             |             |             |             | J | 0,110 | 0,164 | 0,298 | 0,528 | 0,841 | 0,988 |
|     |       |       |             |             |             |             |             | N | 0,083 | 0,127 | 0,246 | 0,460 | 0,790 | 0,961 |
| 5   | 8     | 16    | 0,1         | 0,1         | 0,1         | 0,1         | 0,1         | Y | 0,087 | 0,132 | 0,233 | 0,451 | 0,770 | 0,906 |
|     |       |       |             |             |             |             |             | J | 0,117 | 0,170 | 0,291 | 0,532 | 0,836 | 0,979 |
|     |       |       |             |             |             |             |             | N | 0,086 | 0,129 | 0,231 | 0,453 | 0,777 | 0,928 |
| 5   | 8     | 16    | 0,1         | 0,1         | 0,1         | 0,1         | 0,1         | B | 0,075 | 0,101 | 0,149 | 0,284 | 0,521 | 0,824 |
|     |       |       |             |             |             |             |             | Y | 0,083 | 0,135 | 0,224 | 0,416 | 0,741 | 0,897 |
|     |       |       |             |             |             |             |             | J | 0,114 | 0,174 | 0,283 | 0,498 | 0,811 | 0,970 |
|     |       |       |             |             |             |             |             | N | 0,072 | 0,118 | 0,204 | 0,392 | 0,727 | 0,894 |

| $p$ | $n_1$ | $n_2$ | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | $\lambda_4$ | $\lambda_5$ |   | 1     | 2     | 4     | 8     | 16    | 32    |
|-----|-------|-------|-------------|-------------|-------------|-------------|-------------|---|-------|-------|-------|-------|-------|-------|
| 5   | 8     | 16    | 0,33        | 0,33        | 0,33        | 0,33        | 0,33        | Y | 0,076 | 0,117 | 0,206 | 0,412 | 0,755 | 0,933 |
|     |       |       |             |             |             |             |             | J | 0,106 | 0,157 | 0,264 | 0,489 | 0,815 | 0,976 |
|     |       |       |             |             |             |             |             | N | 0,078 | 0,119 | 0,211 | 0,420 | 0,762 | 0,946 |
|     |       |       |             |             |             |             |             | B | 0,067 | 0,085 | 0,124 | 0,207 | 0,376 | 0,651 |
| 5   | 12    | 12    | 0,1         | 0,1         | 0,1         | 0,1         | 0,1         | Y | 0,102 | 0,138 | 0,241 | 0,402 | 0,694 | 0,805 |
|     |       |       |             |             |             |             |             | J | 0,137 | 0,186 | 0,312 | 0,507 | 0,804 | 0,934 |
|     |       |       |             |             |             |             |             | N | 0,064 | 0,093 | 0,181 | 0,331 | 0,641 | 0,756 |
|     |       |       |             |             |             |             |             | B | 0,076 | 0,106 | 0,172 | 0,319 | 0,596 | 0,899 |

| $p$ | $n_1$ | $n_2$ | $\lambda_1$ | $\lambda_2$ | $\dots$ | $\lambda_D$ |   | 1     | 2     | 4     | 8     | 16    | 32    |
|-----|-------|-------|-------------|-------------|---------|-------------|---|-------|-------|-------|-------|-------|-------|
| 10  | 15    | 15    | 0,5         | 0,5         | $\dots$ | 0,5         | Y | 0,060 | 0,084 | 0,134 | 0,263 | 0,550 | 0,696 |
|     |       |       |             |             |         |             | J | 0,136 | 0,179 | 0,258 | 0,436 | 0,736 | 0,926 |
|     |       |       |             |             |         |             | N | 0,064 | 0,089 | 0,141 | 0,274 | 0,568 | 0,727 |
|     |       |       |             |             |         |             | B | 0,067 | 0,085 | 0,127 | 0,227 | 0,450 | 0,793 |
| 10  | 15    | 30    | 0,33        | 0,33        | $\dots$ | 0,33        | Y | 0,068 | 0,097 | 0,162 | 0,327 | 0,654 | 0,898 |
|     |       |       |             |             |         |             | J | 0,102 | 0,140 | 0,222 | 0,410 | 0,732 | 0,956 |
|     |       |       |             |             |         |             | N | 0,070 | 0,100 | 0,166 | 0,332 | 0,660 | 0,904 |
|     |       |       |             |             |         |             | B | 0,062 | 0,074 | 0,102 | 0,164 | 0,304 | 0,572 |

A few of these power functions are sketched in the following figures:  
 (James: - · - · ; NAV: ———; Yao: ---; Bennett: .....)





## REFERENCES

- ANDERSON, T.W. (1958). An introduction to multivariate statistical analysis. Wiley, New York.
- BEHRENS, W.U. (1929). Ein Beitrag zur Fehlerberechnung bei wenigen Beobachtungen. *Landw. Jb.* 68, 807–837.
- BENNETT, B.M. (1951). Note on a solution of the generalized Behrens–Fisher problem. *Ann. Inst. Statist. Math.* 2, 87–90.
- DE LA REY, N. (1987). A comparison of several solutions to the multivariate Behrens–Fisher problem. *Tech. Report No. 136*, Dept. of Math. Statist., University of the Orange Free State.
- FISHER, R.A. (1935). The fiducial argument in statistical inference. *Ann. of Eugenics* 6, 395–398.
- JAMES, G.S. (1954). Tests of linear hypotheses in univariate and multivariate analysis when the ratios of the population variances are unknown. *Biometrika* 41, 19–43.
- NEL, D.G. & VAN DER MERWE, C.A. (1986). A solution to the multivariate Behrens–Fisher problem. *Commun. Statist. – Theory Meth.* 15 (12), 3719–3735.
- SCHEFFÉ, H. (1943). On solutions of the Behrens–Fisher problem, based on the *t*-distribution. *Ann. Math. Statist.* 10, 129–137.
- SUBRAHMANIAM, K. & SUBRAHMANIAM, K. (1973). On the multivariate Behrens–Fisher problem. *Biometrika* 60, 107–111.
- TAN, W.Y. & GUPTA, R.P. (1983). On approximating a linear combination of central Wishart matrices with positive coefficients. *Comm. Statist. Theor. Meth.* 12(22), 2589–2600.
- WELCH, B.L. (1936). Specification of rules for rejecting too variable a product, with particular reference to an electric lamp problem. *J.R. Statist. Soc. Supp.* 3, 29–48.
- WELCH, B.L. (1947). The generalization of “Student’s” problem when several different population variances are involved. *Biometrika* 34, 28–35.
- YAO, Y. (1965). An approximate degrees of freedom solution to the multivariate Behrens–Fisher problem. *Biometrika* 52, 139–147.